Host-Galaxy Identification of Continuous-Gravitational-Wave Sources with Low-Frequency VBLI

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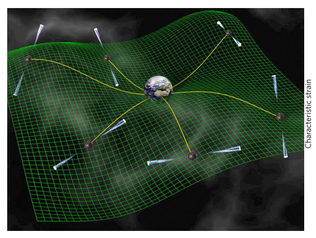
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Introduction

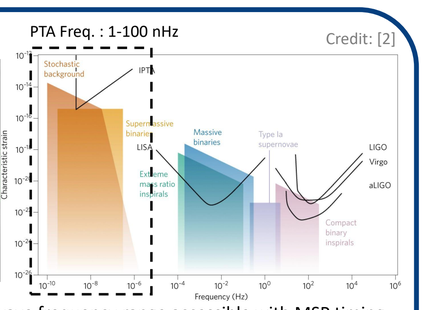
Pulsar Timing Array (PTA)

PTA detects nHz gravitational waves using millisecond pulsars(MSPs) as precise clocks.

- Examples of nHz GWs : GW Background, CGW from SMBHB, …



Credit: [1]



Credit: [2]

Fig. 1. PTA concept and gravitational-wave frequency range accessible with MSP timing

PTA CGW Localization & VLBI Astrometry

Precise VLBI astrometry ⇒ PTA CGW Localization

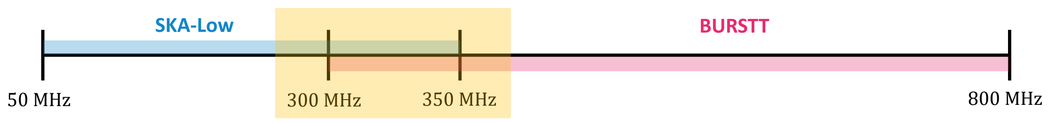
Pulsar-term phase couples source direction and pulsar distance:

$$\varphi_p = 2\pi f_{GW} \frac{L_p}{c}(1 + \hat{\Omega} \cdot \hat{p})$$

Why Low-Frequency VLBI

Pulsars are brighter at low ν — Ionospheric error increases

Expected Crossover
$\nu \sim 300 - 350$ MHz



* A few high-precision anchor pulsars can greatly improve localization.

Motivation : Find new anchor pulsars in low-frequency bands

Method & Setup

Signal & Fisher Localization

- Circular, quasi-monochromatic SMBHB CGW source.
- Residual model includes Earth term and pulsar term.
- Gaussian white timing noise
- Parameters: $\mathbf{x} = \left(\log_{10} M_{\rm chirp}, \log_{10} f_{GW}, \Phi_0, \log_{10} d_L, \iota, \psi, \theta, \phi, L_1, \ldots, L_{N_p}\right)$
- Posterior precision:

$$r_i(t) = \sum_A F_i^A [s_A(t) - s_A(t - \tau_i)]$$

$$P = H + \Lambda_{\rm prior}, \qquad \Lambda_{\rm prior} = \mathrm{diag}\left(0, \cdots, 0, \sigma_1^{-2}, \cdots \sigma_{N_p}^{-2}\right), \qquad H = J^T C^{-1} J$$

VLBI Astrometry Forecast

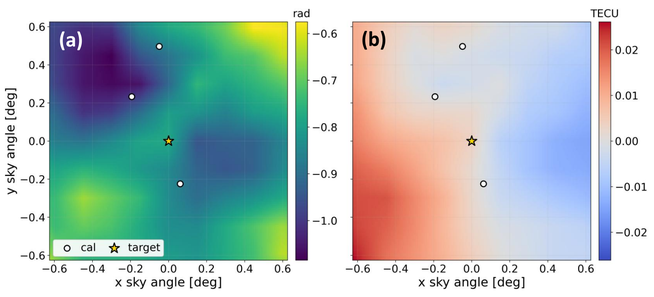


Fig. 2. Example ionospheric field and calibration residual.
(a) Simulated true ionospheric phase screen across the target field.
(b) Residual ΔTEC after MultiView calibration with three calibrators.
Open circles indicate calibrators, and the star indicates the target.

Total Electron Content (TEC)
- phase errors occurred by TEC deviation
- **ΔTEC scaling**: residual ionospheric error is reduced by factors ×1, ×2, ×5 to test the calibration level required for competitive low-frequency VLBI.

- Astrometry uncertainty σ_π is calculated from aperture synthesis with $\sigma_{th}, \sigma_{iono}$.
  - 8 parallax epochs over 2 yr
  - 8 h integration per epoch
  - LAMBDA baseline configuration

* Presentation about LAMBDA : 10.15 ㅜAT-04 (Jiwon Jang)

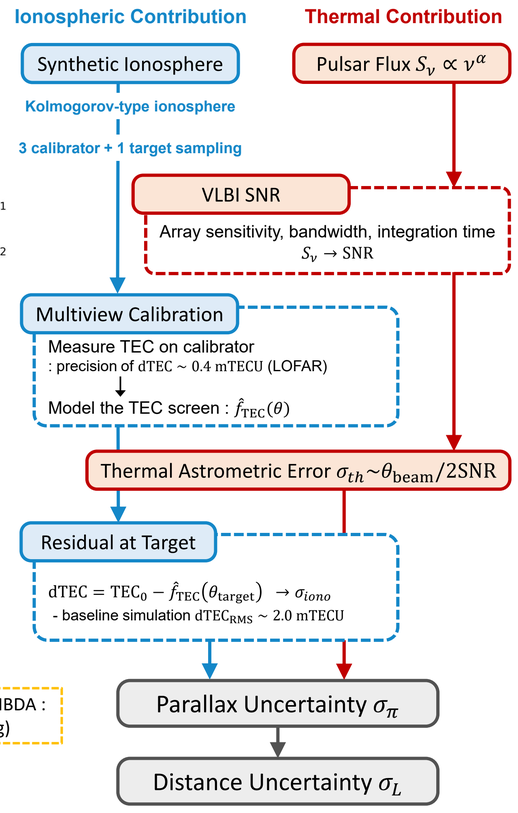


Fig. 3. Ionospheric-error pipeline for the VLBI astrometry forecast.
A synthetic Kolmogorov-type TEC screen is sampled at three calibrators, fitted with MultiView calibration, and evaluated at the target to obtain the residual ΔTEC and $\sigma_{\rm iono}$.

PTA Pulsar Set Construction

: 16-pulsar illustrative PTA sample
- we used ATNF pulsar catalogue
- 8 low thermal noise MSPs, at $\nu = 300$ MHz
- 8 nearby MSPs
- prior Λ with forecasted $\{\sigma_L\}$

Results

Low-Frequency Anchor-Pulsar Candidates

300 MHz observations recover several promising anchor-pulsar candidates despite stronger ionospheric errors.

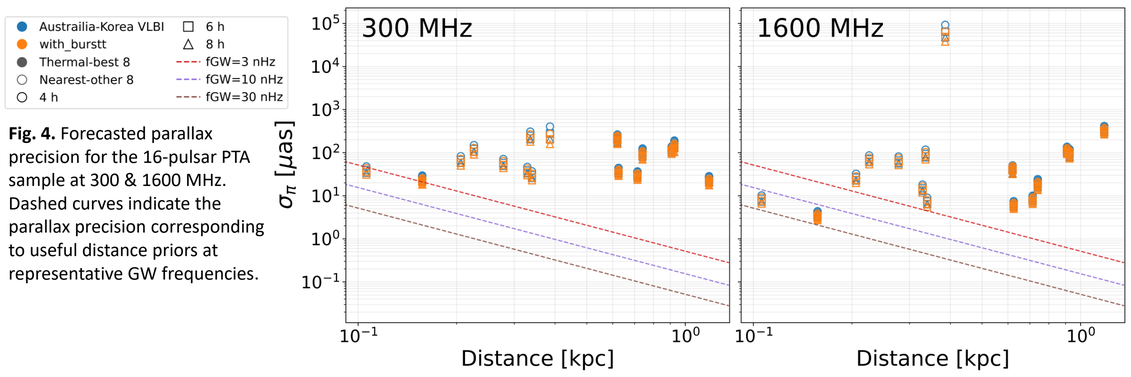


Fig. 4. Forecasted parallax precision for the 16-pulsar PTA sample at 300 & 1600 MHz. Dashed curves indicate the parallax precision corresponding to useful distance priors at representative GW frequencies.

Localization Area

We scale the residual TEC to determine the ionospheric-calibration level required for 300 MHz VLBI to become competitive with the 1600 MHz reference.

* contour shows HECATEv2 galaxy density ($d \leq 200$ Mpc)

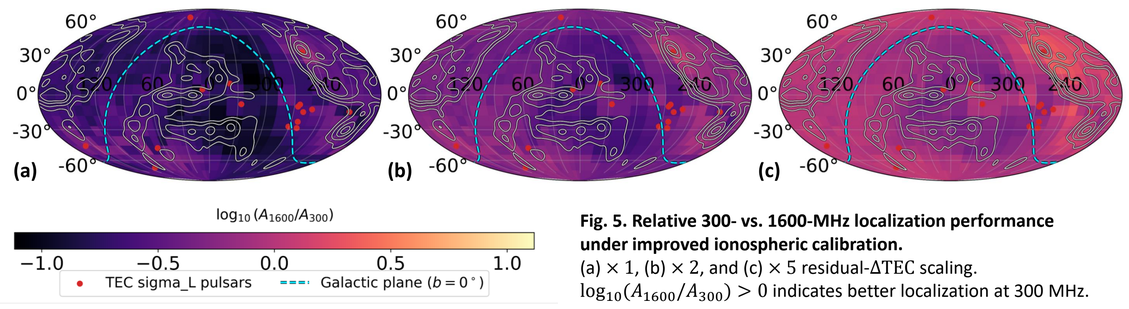


Fig. 5. Relative 300- vs. 1600-MHz localization performance under improved ionospheric calibration.
(a) × 1, (b) × 2, and (c) × 5 residual-ΔTEC scaling.
$\log_{10}(A_{1600}/A_{300}) > 0$ indicates better localization at 300 MHz.

Number of Host-Candidates

Core: ~50 hosts at 300 MHz (x1) → ~25 with x2 calibration, comparable to ~20-30 at 1600 MHz.

- The Hercules cluster(Abell 2151) is particularly favorable for the 300 MHz setup.

* contour shows GLADE+ galaxy density ($d \leq 800$ Mpc)

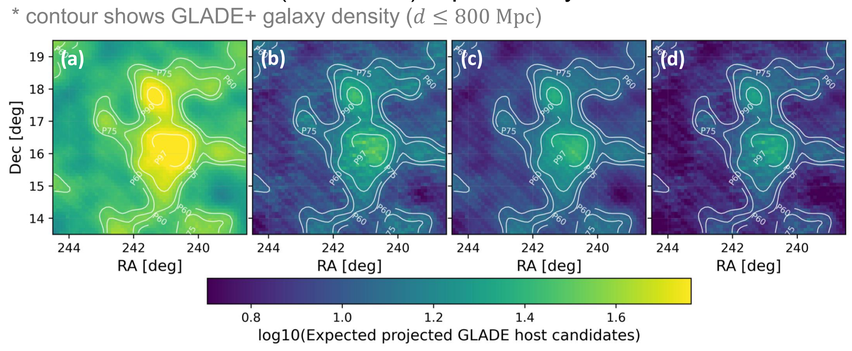


Fig. 6. Expected host-galaxy candidate counts around the Hercules cluster (Abell 2151).
(a) 300 MHz with baseline ionospheric calibration (× 1).
(b) 1600 MHz with baseline calibration (× 1).
(c) 300 MHz with residual-ΔTEC reduced by a factor of 2 (× 2).
(d) 1600 MHz with the same (× 2) calibration scaling.
Colors show $\log_{10}$ of the expected projected GLADE+ host-candidate count; white contours trace the local galaxy-density distribution.

Localization in GW Parameter Space

[For scale]
$M_{chirp} \sim 10^8 - 10^9 M_\odot$: massive SMBHBs
$f_{GW} \sim 1 - 10$ nHz : multi-year orbital timescales

- Low-frequency VLBI astrometry enables a parameter-space forecast of PTA CGW localization performance.
- Low-frequency VLBI retains a relative localization advantage in selected weak-signal

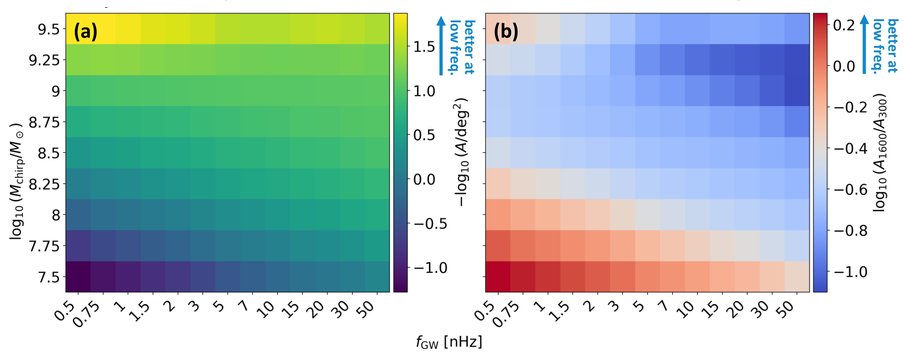


Fig. 7. CGW localization performance across SMBHB parameter space.
(a) Median 300-MHz localization performance, shown as $-\log_{10} A_{300}$. Larger values indicate tighter localization.
(b) Relative performance of 300 and 1600 MHz, shown as $\log_{10}(A_{1600}/A_{300})$. Positive values indicate tighter localization at 300 MHz, while positive values favor 1600 MHz.

Identifying Low-Frequency Anchor Pulsars

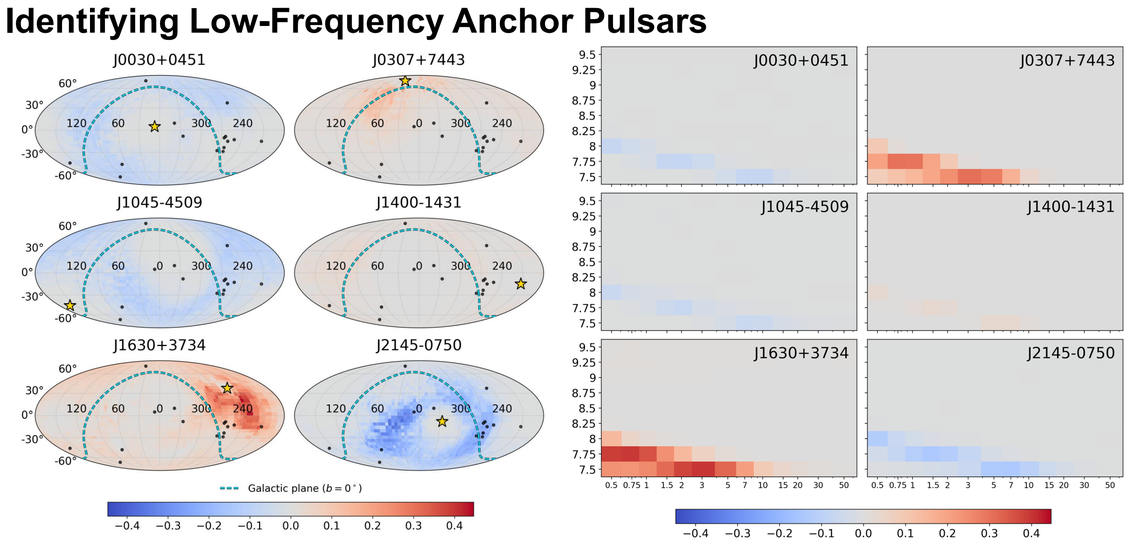


Fig. 8. Sky-dependent localization gain from replacing one pulsar's 1600-MHz astrometric prior with its 300-MHz forecast.
For pulsar i, the gain is quantified as

$$\Delta_i = \log_{10}\left(\frac{A_{1600,\rm all}}{A_{\rm hybrid},i}\right)$$

where $A_{{\rm hybrid},i}$ is the localization area when only pulsar i uses the 300-MHz prior. Positive values indicate improved localization.

J1630+3734 and **J0307+7443** improve CGW localization when observed at 300 MHz.

Fig. 9. Contribution of each pulsar to the fraction of parameter space where 300-MHz astrometry outperforms the 1600-MHz reference.
The leave-one-out contribution is

$$D_i = F_{\rm low}^{\rm all} - F_{\rm low}^{(-i)}, \qquad F_{\rm low} = P(A_{300} < A_{1600})$$

Positive D_i identifies pulsars that strengthen the low-frequency advantage.

J1630+3734 and **J0307+7443** enhance the relative advantage of low-frequency VLBI, particularly in the weak-GW regime.

Conclusion

1. SKA-Low-assisted VLBI astrometry shows potential to improve PTA continuous gravitational-wave (CGW) localization.

2. Low-frequency VLBI remains competitive with mid-frequency VLBI in selected **weak-GW regimes**.

3. High-impact pulsars are identified, suggesting targeted VLBI observations as an effective strategy.

Caveats & Future Work

- Localization performance depends strongly on ionospheric calibration accuracy. - More realistic time-dependent ionospheric modeling and observational validation are required.
- Extended parameter-space analysis and optimized pulsar selection will further assess the potential of SKA-Low VLBI.

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Acknowledgement: This work is supported by the KAIST Undergraduate Research Participation(URP) Program, advised by Prof. Junhan Kim